

Problem

Design a problem to help other students better understand how to determine the rms value of the sum of multiple currents.

Step-by-step solution

Step 1 of 3

Let us consider the sum of multiple currents as

$$i(t) = i_1(t) + i_2(t) + i_3(t)$$

Where

$$i_1(t) = a_1 \text{ A}$$

$$i_2(t) = a_2 \sin(b_1 t + \alpha^\circ) \text{ A}$$

$$i_3(t) = a_3 \cos(b_2 t + \beta^\circ) \text{ A}$$

Step 2 of 3

The rms value of $i_1(t)$ is

$$I_{1\text{rms}} = a_1$$

The rms value of $i_2(t)$ is

$$I_{2\text{rms}} = \frac{a_2}{\sqrt{2}}$$

The rms value of $i_3(t)$ is

$$I_{3\text{rms}} = \frac{a_3}{\sqrt{2}}$$

Step 3 of 3

The sum of multiple currents is

$$i(t) = a_1 + a_2 \sin(b_1 t + \alpha^\circ) + a_3 \cos(b_2 t + \beta^\circ)$$

Thus, the rms value of $i(t)$ is

$$I_{\text{rms}} = \sqrt{I_{1\text{rms}}^2 + I_{2\text{rms}}^2 + I_{3\text{rms}}^2}$$

$$I_{\text{rms}} = \sqrt{(a_1)^2 + \left(\frac{a_2}{\sqrt{2}}\right)^2 + \left(\frac{a_3}{\sqrt{2}}\right)^2}$$

$$\therefore I_{\text{rms}} = \sqrt{a_1^2 + \frac{a_2^2}{2} + \frac{a_3^2}{2}} \text{ A}$$